

11 Quantum states, level densities and level numbers of composite quantum systems

11.1 Harmonic oscillator

$$E_v = h\nu(v + \frac{1}{2}), \quad v = 0, 1, 2, 3, \dots, \infty \quad (11.1)$$

Number of states with energy $E_v < E$ given by

$$W(E) = \sum_{v=0}^{\infty} h(E - E_v), \quad (11.2)$$

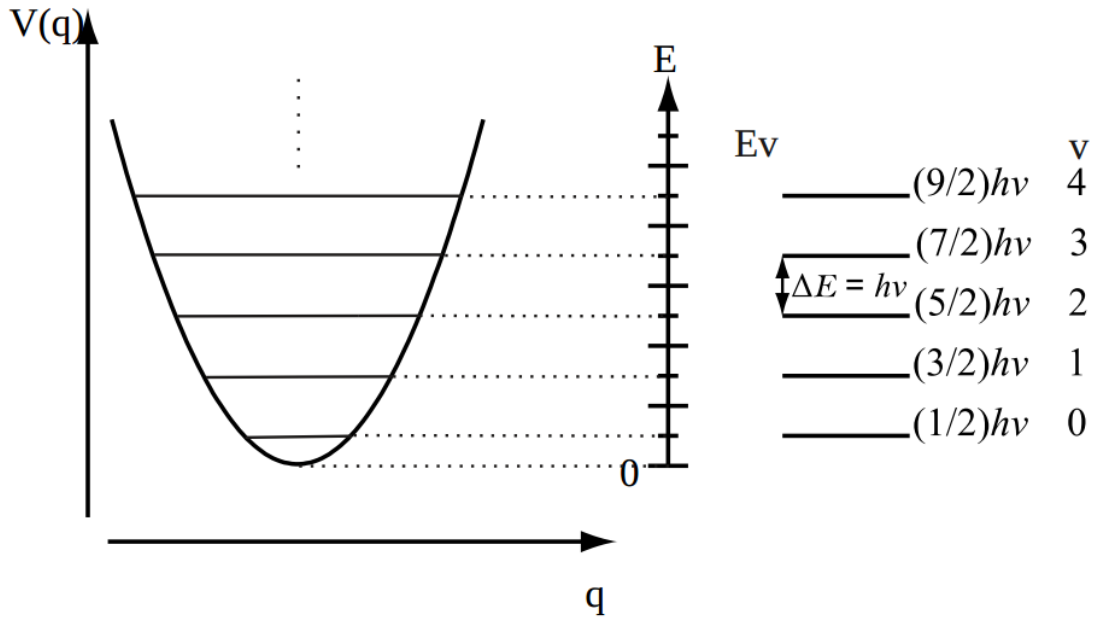


Figure 8: Potential energy $V(q)$ and energy eigenvalues E_v of the harmonic oscillator.

where $h(x)$ is the Heaviside step function defined as

$$\begin{aligned} h(x) &= 0 & \text{for } x < 0 \\ h(x) &= 1 & \text{for } x > 0 \end{aligned} \quad (11.3)$$

Density of states is defined as

$$\rho(E) = \frac{dW(E)}{dE} = \sum_{v=0}^{\infty} \delta(E - E_v) \quad (11.4)$$

Instead of $W(E)$, one often works with the smoothed version $\langle W(E) \rangle$ which describes the averaged behaviour

$$\langle W(E) \rangle = v + 1 = \frac{E}{h\nu} + \frac{1}{2} = \frac{E + E_z}{h\nu} \quad (11.5)$$

The deviation of $\langle W(E) \rangle$ with respect to energy provides the averaged level density:

$$\langle \rho(E) \rangle = \frac{d \langle W(E) \rangle}{dE} = \frac{1}{h\nu} \simeq \frac{\Delta W(E)}{\Delta E} \quad (11.6)$$

Both averaged quantities are only meaningful in the limit of many quantum states per energy interval.

11.2 Many harmonic oscillators

The levels of a quantum system composed of 2 subsystems are given by

$$E_m = E_{n_1, n_2} = E_{n_1} + E_{n_2} \quad (11.7)$$

For example, we consider 2 harmonic oscillators

$$E_k = E_{v_1, v_2} = (v_1 + \frac{1}{2})h\nu_1 + (v_2 + \frac{1}{2})h\nu_2 \quad (11.8)$$

Expressed in the level densities, we have

$$\begin{aligned} N_1(E_1) &= \rho_1(E_1) |dx| \equiv \rho_1(E') |dE'| \\ N_2(E_2) &= \rho_2(E_2) |dx| \equiv \rho_2(E'') |dE''| \end{aligned} \quad (11.9)$$

The total number of levels that fulfill $E_1 + E_2 = E$ ($|dx| \ll E$) is given by

$$N_{1,2} = N_1 N_2 = \rho_1(E_1) \rho_2(E_2) (dx)^2 \quad (11.10)$$

The corresponding level density for $E_1 = x$ and $E_2 = E - x$ is

$$\begin{aligned} \rho_{1,2}(E_1 = x, E_2 = E - x, E) &= \frac{dN_{1,2}}{dx} \\ &= \rho_1(x) \rho_2(E - x) dx \end{aligned} \quad (11.11)$$

The total level density at energy E is then obtained by integrating over the variable x

$$\begin{aligned} \rho_{1,2}(E) &= \int_{x=0}^{x=E} \rho_1(x) \rho_2(E - x) dx \\ &= \int_{E'=0}^{E'=E} \rho_1(E') \rho_2(E'') dE'', \end{aligned} \quad (11.12)$$

which corresponds to the convolution of the functions ρ_1 and ρ_2 , which is also written

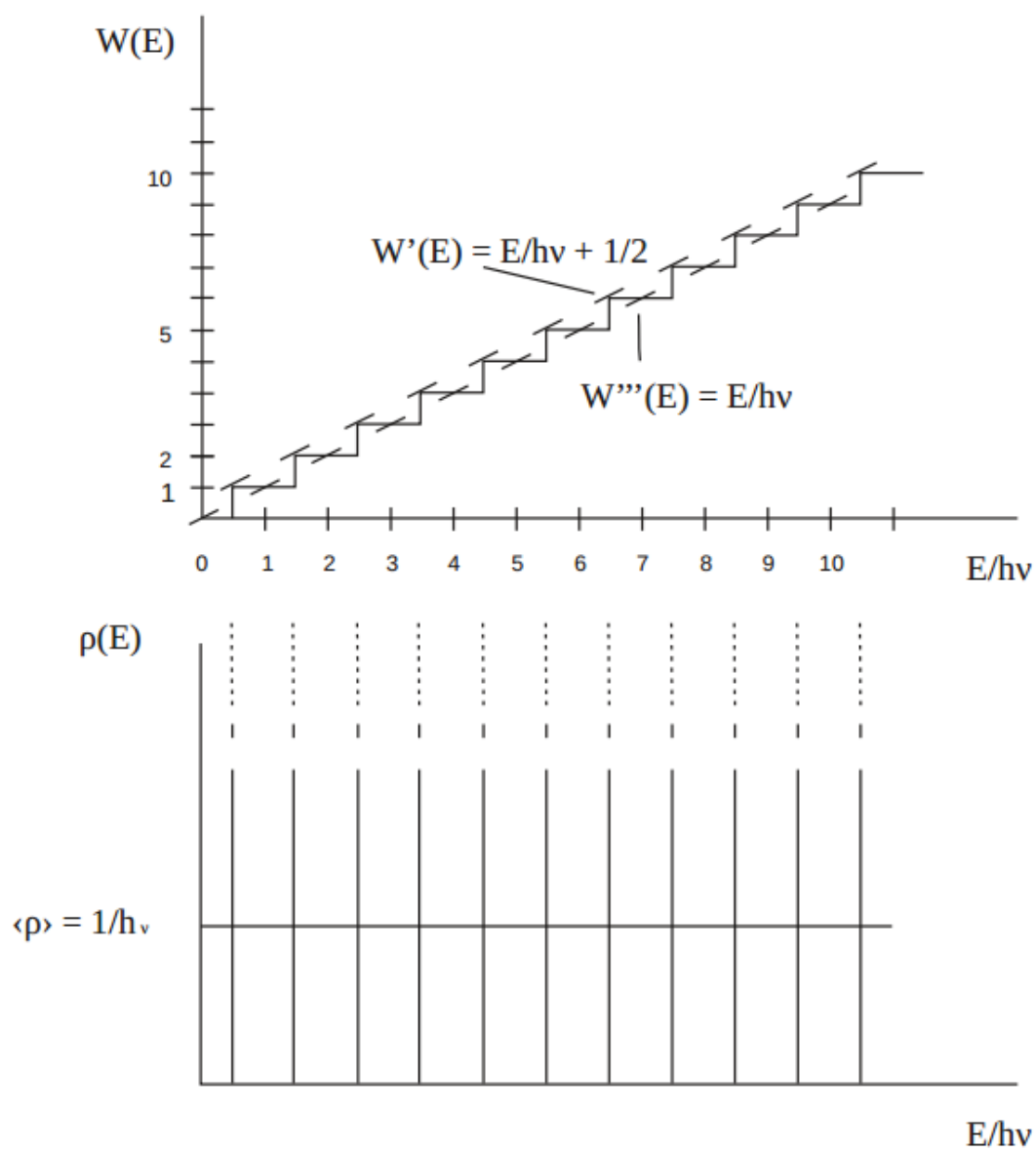


Figure 9: Number of states $W(E)$ and the density of states $\rho(E)$ for a harmonic oscillator.

$$\rho_{1,2} = \rho_1 \otimes \rho_2 = \rho_2 \otimes \rho_1 \quad (11.13)$$

Similarly, the number of levels can also be obtained by convolution

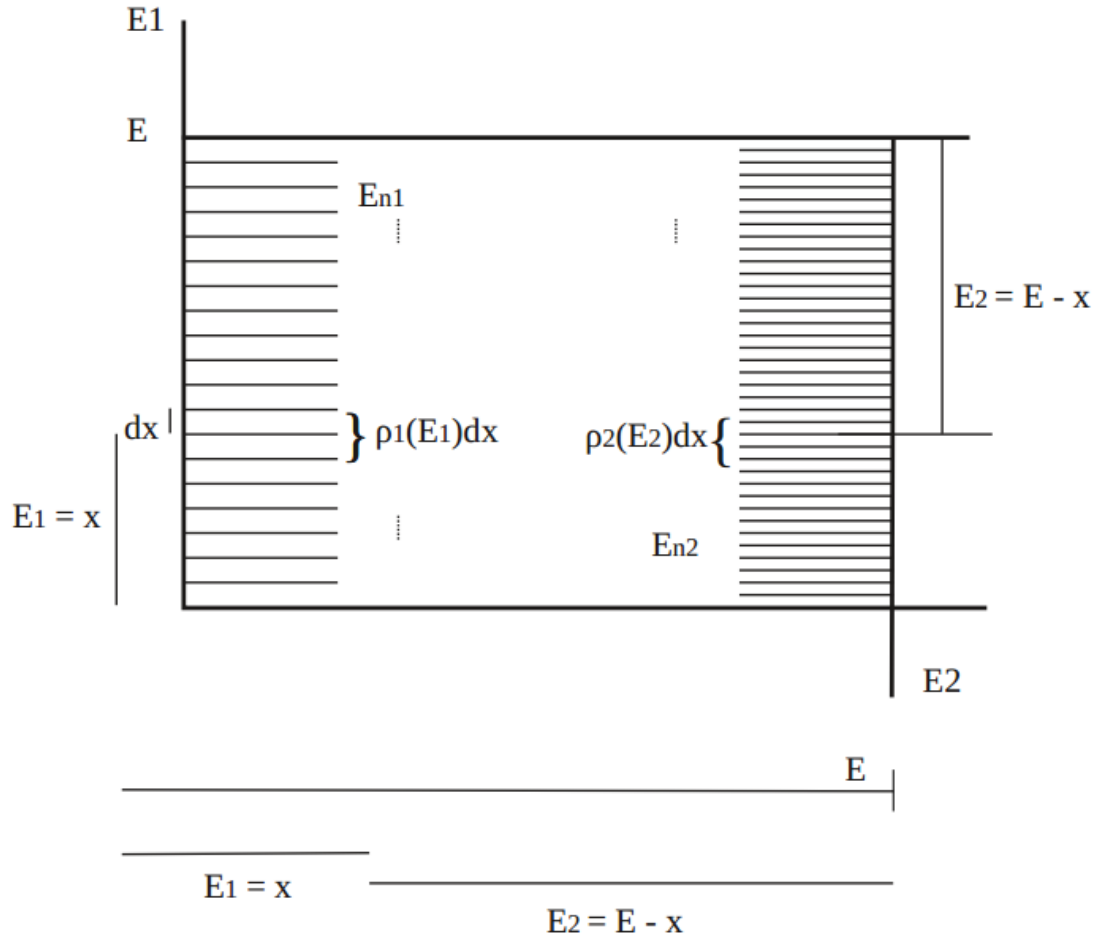


Figure 10: Convolution of the densities of states of two spectra $E_1 + E_2 = E$, see text).

$$\begin{aligned}
 W_{1,2}(E) &= \int_0^E W_1(E') \rho_2(E - E') dE' \\
 &= W_1 \otimes \rho_2 = \rho_1 \otimes W_2 \\
 &= \int_0^E \rho_{1,2}(x) dx
 \end{aligned}
 \tag{11.14}$$

The level density of a system composed of n subsystems can be obtained by a sequence of convolutions

$$\begin{aligned}
\rho_{1,2,3}(E) &= \int_0^E \rho_{1,2}(x) \rho_3(E-x) dx \\
&= \int_0^E \rho_{1,2}(E-E''') \rho_3(E''') dE'''
\end{aligned} \tag{11.15}$$

The application of these formula to the smoothed level density of harmonic oscillators is particularly simple because the level densities are not energy dependent:

$$\begin{aligned}
\rho_{1,2,3} &= (h\nu_1)^{-1} (h\nu_2)^{-1} (h\nu_3)^{-1} \int_0^E E''' dE''' \\
&= (h\nu_1 h\nu_2 h\nu_3)^{-1} \frac{1}{2} E^2
\end{aligned} \tag{11.16}$$

For s harmonic oscillators we therefore find

$$\rho_{1,2,3,\dots,s}(E) = \frac{E^{s-1}}{(s-1)! \prod_{i=1}^s h\nu_i} \tag{11.17}$$

The number of levels is obtained by integration

$$\begin{aligned}
W_{1,2,3,\dots,s} &= \frac{E^s}{s! \prod_{i=1}^s h\nu_i} \\
&= \int_0^E \rho_{1,2,3,\dots,s}(E'''\dots') dE'''\dots'
\end{aligned} \tag{11.18}$$

These simple formulae represent an approximate result following from the use of the smoothed density function $\langle \rho(E) \rangle \approx (h\nu_i)^{-1}$. A more accurate calculation provides the following better approximation

$$\begin{aligned}
\rho_{1,2,3,\dots,s} &= \frac{(E + aE_z)^{s-1}}{(s-1)! \prod_{i=1}^s (h\nu_i)} \\
W_{1,2,3,\dots,s} &= \frac{(E + aE_z)^s}{s! \prod_{i=1}^s (h\nu_i)}
\end{aligned} \tag{11.19}$$

where E_z is the total zero-point energy of the s harmonic oscillators

$$E_z = \sum_{i=1}^s \frac{1}{2} (h\nu_i) \tag{11.20}$$

and a is an energy-dependent parameter which can be chosen as $a = 1$ for the purpose of a simple approximation. This yields a upper boundary for $\rho(E)$ whereas $a = 0$ represent a lower boundary. $a = 1$ is often called the "semi-classical approximation", whereas $a = 0$ is the "classical-approximation".

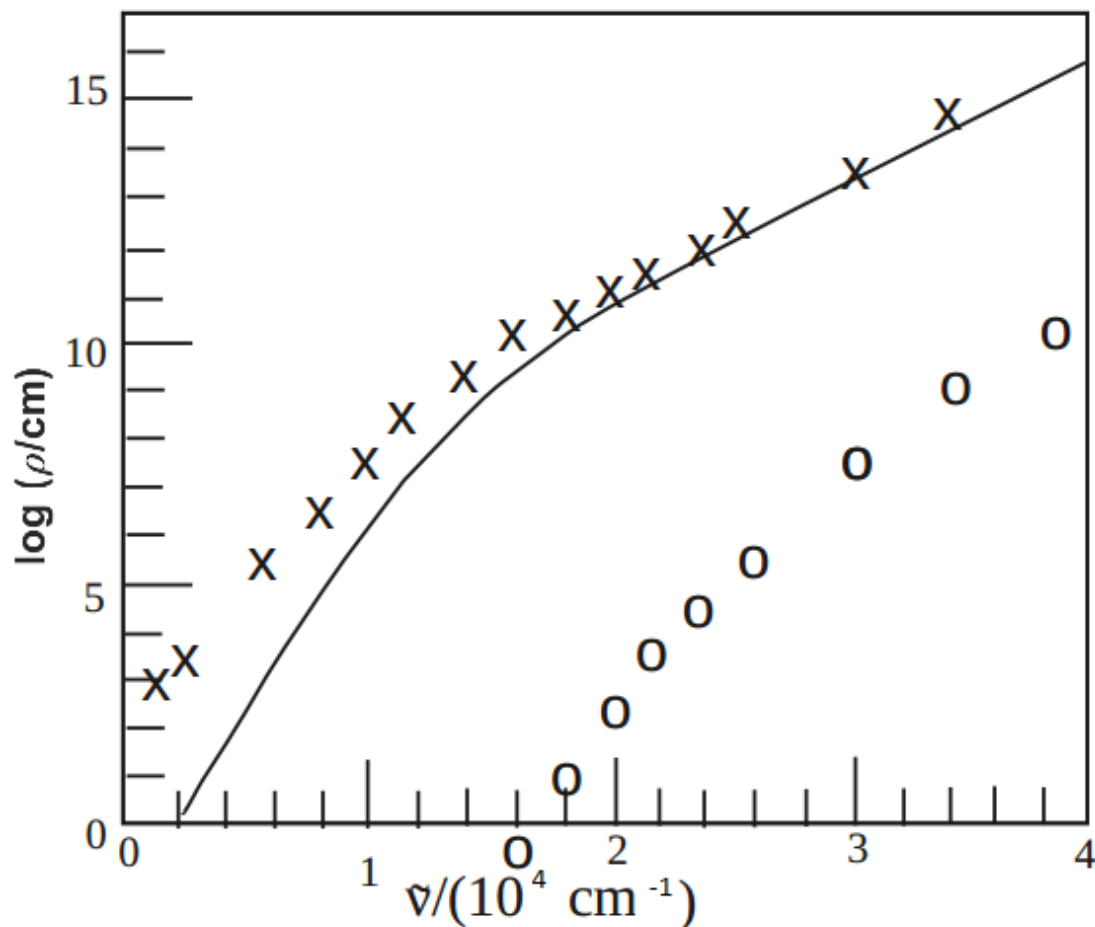


Figure 11: Logarithm (base 10) of the density of states for 30 vibrational degrees of freedom in $\text{C}_6\text{H}_5\text{F}$, Symbol x="semiclassical approximation" with $a = 1$ in Equation.(11.19); Symbol o="classical approximation". The solid line results from a step function in quantum mechanics "exact" step function $W(E)$, (from Chem.Phys. 51,353 (1980)). Wave number units have been used, which are related to energy by $E = hc\tilde{\nu}$ for the density of states $\rho(E) = \frac{\tilde{\rho}}{hc}$.

11.3 Many particles in a box (model for ideal gas)

Energy levels of a particle of mass m in one-dimensional box of length a are given by

$$E_n = \frac{n^2 h^2}{8ma^2}, \quad n = 1, 2, 3, \dots \quad (11.21)$$

The quantum number n counts the number of levels, hence

$$W(E) = n = \frac{2a}{h} \sqrt{2mE} \quad (11.22)$$

The level density is obtained through derivation

$$\rho(E) = \frac{dW(E)}{dE} = \frac{2a}{h} \sqrt{2m} \frac{1}{2} E^{-\frac{1}{2}} = \frac{a}{h} \sqrt{2m} E^{-\frac{1}{2}} \quad (11.23)$$

The level density for a particle in a three-dimensional box is obtained by convolution

$$\rho_d = \rho_{x,y,z} = \rho_x \otimes \rho_y \otimes \rho_z = m^{\frac{3}{2}} 2^{\frac{5}{2}} \pi a^3 h^{-3} E^{\frac{1}{2}} \quad (11.24)$$

In contrast to the spectrum of the harmonic oscillator, the level densities for a particle in a box (1D or 3D) depend on the energy. Explicit integration yields,

$$W_d(E) = \int \rho_d(E) dE = \pi V h^{-3} m^{\frac{3}{2}} \frac{2^{\frac{7}{2}}}{3} E^{\frac{3}{2}} \quad (11.25)$$

One can again show that a slightly better approximation of the step-like level-number function is possible:

$$\begin{aligned} \ln[W_d(x)] &= \frac{3}{2}x - ce^{-bx} + h\left(\frac{\pi}{6}\right) \\ x &= \ln(8ma^2 \frac{E}{h^2}) \end{aligned} \quad (11.26)$$

For N particles of the same mass m in a box we find

$$\rho_N^{(E)} = \rho_{d_1} \otimes \rho_{d_2} \otimes \rho_{d_3} \otimes \dots \otimes \rho_{d_N} \quad (11.27)$$

The corresponding number of states is

$$\begin{aligned} W_N(E) &= \int_0^{(E)} \rho_N(x) dx = \alpha E^{\frac{3N}{2}} \\ \text{where } \alpha &= [2\pi V \frac{(2m)^{\frac{3}{2}}}{h^3}]^N \frac{[\Gamma(\frac{3}{2})]^N}{\Gamma[\frac{3}{2}N + 1]} \end{aligned} \quad (11.28)$$

where a^3 is the volume V of the box and we call the properties of the gamma function

$$\begin{aligned} \Gamma(\frac{3}{2}) &= \frac{\sqrt{\pi}}{2} \\ \Gamma(n) &= (n-1)! \quad \text{for } n = 1, 2, 3, \dots \end{aligned} \quad (11.29)$$

In the case of N identical, non-distinguishable particle, such as gas atoms of the same isotope, we need to take into account the Pauli principle, i.e. the fact that only a function $(N!)^{-1}$ of these states is allowed. Therefore

$$W_{N,\text{Pauli}}(E) = \alpha (N!)^{-1} E^{\frac{3N}{2}} \quad (11.30)$$

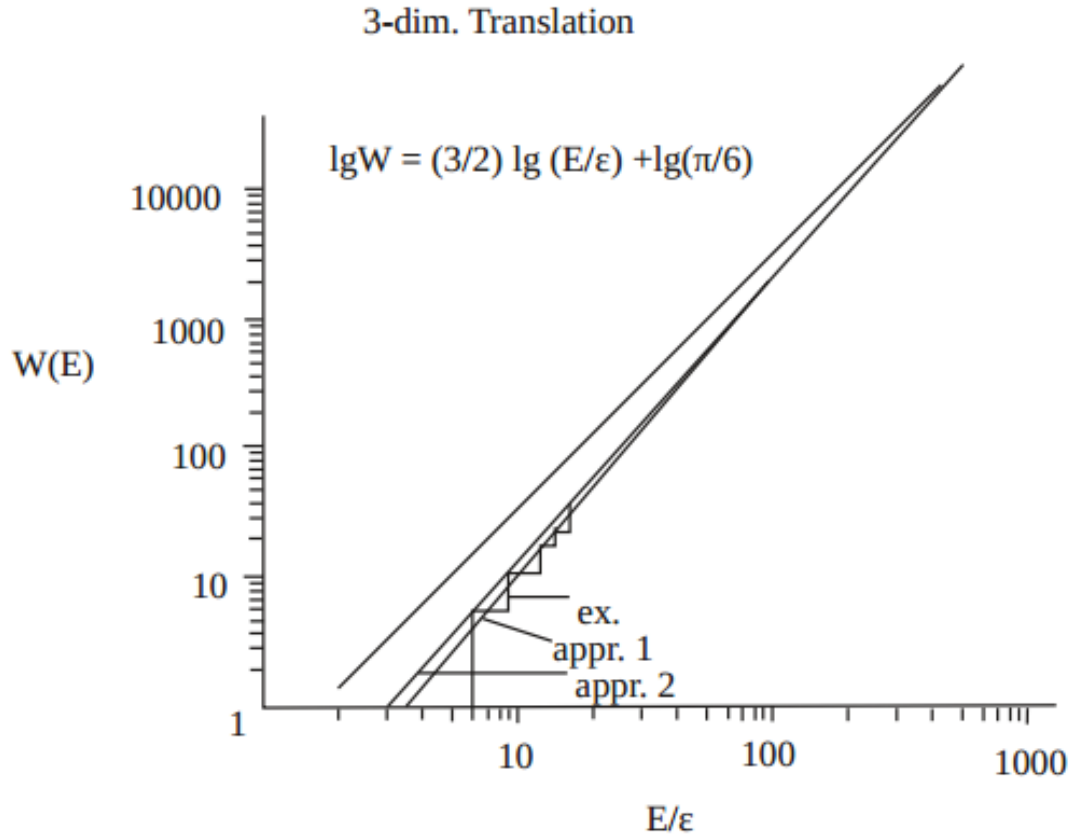


Figure 12: Partition function $W(E)$ for the three-dimensional translational motion of a particle in the box, exact step function and continuous approximations (appr. 1, appr.2 (logarithmic scale)). The approximation 2 (appr.2) has the particular advantage that $W(E) = 1$ for $E = E_z(3\epsilon, \text{Zero-point energy})$ and is of the form in Eq. (11.26) with $c = [1.5 \ln 3 + \ln(\frac{\pi}{6})]e^{b \ln 3}$, where $b = 0.413$ is a good approximation (variable). The approximation 1 corresponds to Eq. (11.26) with $c = 2.35, b = 0.48, \epsilon = \frac{h^2}{8ma^2}$.

This simple correction by division through $N!$ is only valid for high energies E and very large values of $W(E)$.

These results obtained so far allowed us to draw the following conclusions about quantum levels and properties of macroscopic systems:

1. The average energy-level separation of neighbouring quantum states is given by the inverse average level density and becomes unmeasurably small for all known forms of spectroscopy.
2. As a consequence of 1, it is experimentally impossible to prepare single quantum levels as stationary energy eigenstates. The **concept** of stationary states therefore loses its usefulness

when mesoscopic or macroscopic systems are considered.

3. Since the experimentally observable quantum states of macroscopic systems are therefore always superposition of many energy levels, they have to always be considered as **time-dependent** states.
4. The vary large numbers involved suggest or require the application of statistical methods.

11.4 Laplace transformation for derivation of level densities and level numbers via partition function

The derivation of the level densities from convolutions can be more accurately obtained via the Laplace transformation. In general, the Laplace transform $g(y)$ of a function $f(x)$ is defined as

$$g(y) = \int_0^\infty f(x)e^{-xy}dx \quad (11.31)$$

In operator notation: $g(y) = \hat{L}[f(x)]$,

The Laplace transformation operator $\hat{L}$ is an integral operator. The inverse transformation with the operator $\hat{L}^{-1}$ corresponds to integration in the complex plane

$$\begin{aligned} f(x) &= \hat{L}^{-1}[g(y)] \\ f(x) &= \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} g(y)e^{xy}dy \end{aligned} \quad (11.32)$$

The following properties are important for our purpose: Convolution theorem: With $F=f_1 \otimes f_2$ the convolution of functions $f_1(x)$ and $f_2(x)$, and $g_1(y) = \hat{L}f_1(x)$, $g_2(y) = \hat{L}f_2(x)$, the function $G = g_1 \cdot g_2 = \hat{L}F$. This means that **convolution** in the function space becomes a simple product in the space of the Laplace transforms. This theorem can now be exploited to obtain a basic access to level densities, provided that the Laplace transform of the level density of the sub-systems is known. We define the partition function (German: Zustandssumme) q as the Laplace transform of the level density

$$\begin{aligned} q(\beta) &= \int_0^\infty \rho(E)e^{-\beta E}dE = \hat{L}[\rho(E)] \\ &= \sum_{j=0}^\infty e^{-\beta E_j} \\ &= \sum_{j=0}^\infty g_j e^{-\beta E_j} \end{aligned} \quad (11.33)$$

where we have used a property of the Dirac δ -function

$$\begin{aligned} \int_{-\infty}^\infty \delta(x - x_0)f(x)dx &= f(x_0) \\ \int_{-\infty}^\infty \delta(E - E_j)e^{-\beta E}dE &= e^{-\beta E_j} \end{aligned} \quad (11.34)$$

and $\beta = (kT)^{-1}$ with the Boltzmann constant k and the temperature T .

We again consider two subsystems with energy levels E_{n_1}, E_{n_2} . The partition functions are

$$\begin{aligned} q_1(\beta) &= \sum_{n_1=0}^{\infty} e^{-\beta E_{n_1}} \\ q_2(\beta) &= \sum_{n_2=0}^{\infty} e^{-\beta E_{n_2}} \end{aligned} \tag{11.35}$$

The partition function of the total system is

$$\begin{aligned} q_{1,2} &= q_1 \cdot q_2 = \left(\sum_{n_1=0}^{\infty} e^{-\beta E_{n_1}} \right) \left(\sum_{n_2=0}^{\infty} e^{-\beta E_{n_2}} \right) \\ &= \sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} e^{-\beta(E_{n_1} + E_{n_2})} \\ &= \sum_{m=0}^{\infty} e^{-\beta E_m} \end{aligned} \tag{11.36}$$

Without the need of approximations, the properties of the Laplace transformation thus lead to the result

$$\begin{aligned} \rho_{1,2} &= \hat{L}^{-1}(q_1 \cdot q_2) \\ &= \hat{L}^{-1} \hat{L}(\rho_1 \otimes \rho_2) \\ &= \rho_1 \otimes \rho_2 \quad q.e.d \end{aligned} \tag{11.37}$$

We now show that the results of section 6.1.3 can be obtained in a more straight-forward way by using the Laplace transform. We start from the translational partition function q_x using a smoothed level density ρ_x ;

$$\begin{aligned} q_x &= \int_0^{\infty} \rho_x e^{-\beta E} dE \\ &= \frac{a}{h} (2m)^{\frac{1}{2}} \int_0^{\infty} E^{-\frac{1}{2}} e^{-\beta E} dE \end{aligned} \tag{11.38}$$

Using

$$\int_0^{\infty} x^n e^{-bx} dx = \frac{\Gamma(n+1)}{b^{n+1}} \tag{11.39}$$

for all real $n > -1$ and all $b > 0$. We further recall that $\Gamma(n+1) = n!$ for all integer n . Here we need $n = -\frac{1}{2}$ and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, therefore

$$q_x = \frac{a}{h} (2m)^{\frac{1}{2}} \frac{\sqrt{\pi}}{\beta^{\frac{1}{2}}} = \left(\frac{2\pi m a^2}{\beta h^2} \right)^{\frac{1}{2}} \tag{11.40}$$

The product is therefore given by

$$q_d = q_x q_y q_z = \left(\frac{2\pi m a^2}{\beta h^2}\right)^{\frac{3}{2}} = V \left(\frac{2\pi m a^2}{\beta h^2}\right)^{\frac{3}{2}} \quad (11.41)$$

Using the following properties of the Laplace transform:

$$\begin{aligned} \hat{L}^{-1}\left(\frac{1}{\beta^n}\right) &= \left(\frac{E^{n-1}}{\Gamma(n)}\right) \\ \text{and } \hat{L}\left(\frac{E^{n-1}}{\Gamma(n)}\right) &= \frac{1}{\beta^n} \end{aligned} \quad (11.42)$$

We obtain

$$\hat{L}^{-1}(q_d) = \rho_d = V \left(\frac{2\pi m}{h^2}\right)^{\frac{3}{2}} \frac{2E^{\frac{1}{2}}}{\sqrt{\pi}}, \quad (11.43)$$

which is identical with Equation (11.24) The partition function for N particles is then a product

$$Q = q_d^N = V^N \left(\frac{2\pi m}{\beta h^2}\right)^{\frac{3N}{2}} \quad (11.44)$$

The inverse of the Laplace transform with $n = \frac{3N}{2}$ provides

$$\begin{aligned} \hat{L}^{-1}(Q) &= \rho_N = V^N \left(\frac{2\pi m}{\beta h^2}\right)^{\frac{3N}{2}} E^{(3N/2-1)} / \Gamma(3N/2) \\ W_N(E) &= \int_0^E \rho_N(x) dx = V^N \left(\frac{2\pi m}{h^2}\right)^{\frac{3N}{2}} \frac{E^{(3N/2)}}{\frac{3N}{2} \Gamma(\frac{3N}{2})} \end{aligned} \quad (11.45)$$

Using the properties $\Gamma(x+1) = x\Gamma(x)$ and $\Gamma(\frac{3}{2}) = \frac{\sqrt{\pi}}{2}$ and leads back to equation(11.28).

11.5 Recurrence times in dense spectra

The high level densities of "dense" spectra typical of large molecules excited to high energies or mesoscopic/macroscopic systems lead to very long recurrence times

$$\tau(E) \geq h \langle \rho(E) \rangle, \quad (11.46)$$

where $\langle \rho(E) \rangle$ is the average level density. The equality holds for an equidistant spectrum ($\langle \rho(E) \rangle = \frac{1}{\delta} = \text{constant}$).

Example:

1. $\text{C}_6\text{H}_5\text{F}$ excited to 18000 cm^{-1} : $\tau_R \geq 0.3 \text{ s}$,
2. $(\text{CF}_3)_3\text{CH}$ excited to 15000 cm^{-1} : $\tau_R \geq 0.3 \times 10^5 \text{ s}$ ($\approx 4 \text{ days}$)
3. 1 mole of Hg(s) with internal energy of 4 kJ: $\tau_R \geq 10^{10^{24}} \text{ s}$. (Longer than the age of the universe: $\tau \approx 1.5 \cdot 10^9 \text{ years} < 10^{18} \text{ s}$).

These results show that the **periodicity** of quantum dynamics is irrelevant for macroscopic systems. This however does not exclude the existence of (seemingly) periodic motion on much shorter time scales (e.g. motion of a pendulum, motion of planets). However, these motions are never really periodic but always damped on longer time scales.

11.6 Unimolecular rate theory

Unimolecular rate theory is best understood using microcanonical rate theory. A molecule must be excited to an internal energy E by collisions or by photoabsorption before reacting, which can either be a rearrangement or breaking into two pieces.

In one dimension, the microcanonical rate is easy to calculate using the flux-side correlation function with a microcanonical distribution:

$$\begin{aligned}\lim_{t \rightarrow 0+} C_{\text{fs}}(t) &= \frac{1}{2\pi\hbar} \iint \delta(x - x^\ddagger) \frac{p}{m} \theta(p) \delta(H - E) dx dp \\ &= \frac{1}{2\pi\hbar} \int_0^\infty \frac{p}{m} \delta\left(\frac{p^2}{2m} + V^\ddagger - E\right) dp \\ &= \frac{1}{2\pi\hbar} \theta(E - V^\ddagger)\end{aligned}$$

We are allowed to take the short-time limit, because in 1D, there is no recrossing as long as the dividing surface is chosen correctly. The microcanonical rate is therefore:

$$k(E) = \lim_{t \rightarrow 0+} \frac{C_{\text{fs}}(t)}{\rho_{\text{r}}(E)} = \frac{\theta(E - V^\ddagger)}{2\pi\hbar \rho_{\text{r}}(E)} = \frac{P_{\text{transmission}}(E)}{2\pi\hbar \rho_{\text{r}}(E)}$$

where we have noted that the Heaviside step function is the classical transmission function discussed in the lecture on scattering theory.

In Rice–Ramsperger–Kassels–Marcus (RRKM) theory, one deals with a multidimensional system in which it is assumed that all reactant states with energy E are equally likely to be populated. If the molecule was excited to a particular mode, this means that we assume that intramolecular vibrational redistribution occurs on a much faster timescale than the reaction. Then we say that there is a reaction mode with energy E_{rxn} , whereas all other modes in states $\mathbf{n} = (n_1, n_2, \dots, n_{f-1})$ have vibrational energy $E_{\mathbf{n}}$, such that the total energy is $E = E_{\text{rxn}} + E_{\mathbf{n}}$. Dynamics along the reaction coordinate is assumed to be classical (so there is no tunnelling) and the reaction can only proceed if $E_{\text{rxn}} > V^\ddagger$. Under these assumptions, the total reaction rate is given by

$$\begin{aligned}\lim_{t \rightarrow 0+} C_{\text{fs}}(t) &= \sum_{\mathbf{n}} \frac{1}{2\pi\hbar} \iint \delta(x - x^\ddagger) \frac{p}{m} \theta(p) \delta(H_{\text{rxn}} + E_{\mathbf{n}}^\ddagger - E) dx dp \\ &= \sum_{\mathbf{n}} \frac{1}{2\pi\hbar} \theta(E_{\text{rxn}} - V^\ddagger) \quad \text{where} \quad E_{\text{rxn}} = E - E_{\mathbf{n}}^\ddagger \\ k(E) &= \lim_{t \rightarrow 0+} \frac{C_{\text{fs}}(t)}{\rho_{\text{r}}(E)} = \frac{N^\ddagger(E)}{2\pi\hbar \rho_{\text{r}}(E)} \quad \text{where} \quad N^\ddagger(E) = \sum_{\mathbf{n}} \theta(E - E_{\mathbf{n}}^\ddagger - V^\ddagger)\end{aligned}$$

It is common to assume that the transition state can be described by a set of harmonic oscillators perpendicular to the reaction coordinate. For small molecules, one can then compute $k(E)$ by summing over $\mathbf{n}$. However, for larger molecules the number of terms in the sum soon becomes so great that one resorts to a classical approximation. This is valid for energies much higher than $V^\ddagger$, where the density of states is large.

$$k_{\text{RRKM}}(E) = \frac{N^\ddagger(E)}{2\pi\hbar \rho_{\text{r}}(E)} \quad \rho_{\text{r}}(E) = \frac{E^{f-1}}{(f-1)! \prod_{j=1}^f \hbar \omega_j}$$

where

$$\begin{aligned}
 N^\ddagger(E) &= \sum_{\mathbf{n}} \theta(E - E_{\mathbf{n}}^\ddagger - V^\ddagger) = \int \sum_{\mathbf{n}} \delta(E' - E_{\mathbf{n}}^\ddagger) \theta(E - E' - V^\ddagger) dE' \\
 &= \int \rho^\ddagger(E') \theta(E - E' - V^\ddagger) dE' \quad \text{where} \quad \rho^\ddagger(E') = \sum_{\mathbf{n}} \delta(E' - E_{\mathbf{n}}^\ddagger) \\
 &= \int_0^\infty \frac{(E')^{f-2}}{(f-2)! \prod_{j=1}^{f-1} \hbar \omega_j^\ddagger} \theta(E - E' - V^\ddagger) dE' \\
 &= \int_0^{E-V^\ddagger} \frac{(E')^{f-2}}{(f-2)! \prod_{j=1}^{f-1} \hbar \omega_j^\ddagger} dE' = \frac{(E - V^\ddagger)^{f-1}}{(f-1)! \prod_{j=1}^{f-1} \hbar \omega_j^\ddagger} \quad E > V^\ddagger
 \end{aligned}$$

so the RRKM rate in this approximation is

$$\begin{aligned}
 k_{\text{RRKM}}(E) &= \frac{N^\ddagger(E)}{2\pi\hbar \rho_r(E)} \\
 &= (2\pi\hbar)^{-1} \frac{(E - V^\ddagger)^{f-1}}{(f-1)! \prod_{j=1}^{f-1} \hbar \omega_j^\ddagger} \frac{(f-1)! \prod_{j=1}^f \hbar \omega_j}{E^{f-1}} \\
 &= (2\pi\hbar)^{-1} \frac{\prod_{j=1}^f \hbar \omega_j}{\prod_{j=1}^{f-1} \hbar \omega_j^\ddagger} \left(\frac{E - V^\ddagger}{E} \right)^{f-1}
 \end{aligned}$$

It is clear that the rate increases with E and decreases for larger molecules (as it is unlikely to find the energy distributed such that there is enough in the reaction coordinate).

From the microcanonical rate, it is easy to get the thermal rate by averging over energies:

$$\text{thermal rate:} \quad k(\beta) = \int_0^\infty k(E) P_{\text{eq}}(E) dE$$

where for a set of classical harmonic oscillators,

$$\begin{aligned}
 P_{\text{eq}}(E) &= \rho_r(E) \frac{e^{-\beta E}}{Q_r(\beta)} \\
 &= \frac{E^{f-1}}{(f-1)! \prod_{j=1}^f \hbar \omega_j} e^{-\beta E} \prod_{j=1}^f \beta \hbar \omega_j \\
 &= \frac{\beta^f E^{f-1}}{(f-1)!} e^{-\beta E}
 \end{aligned}$$

The thermal rate is therefore

$$\begin{aligned}
 k(\beta) &= \int k_{\text{RRKM}}(E) P_{\text{eq}}(E) \, \mathrm{d}E \\
 &= \int_{V^\ddagger}^{\infty} (2\pi\hbar)^{-1} \frac{\prod_{j=1}^f \hbar\omega_j}{\prod_{j=1}^{f-1} \hbar\omega_j^\ddagger} \left(\frac{E - V^\ddagger}{E} \right)^{f-1} \frac{\beta^f E^{f-1}}{(f-1)!} e^{-\beta E} \, \mathrm{d}E \\
 &= \int_{V^\ddagger}^{\infty} (2\pi\hbar)^{-1} \frac{\prod_{j=1}^f \hbar\omega_j}{\prod_{j=1}^{f-1} \hbar\omega_j^\ddagger} (E - V^\ddagger)^{f-1} \frac{\beta^f}{(f-1)!} e^{-\beta E} \, \mathrm{d}E \\
 &= (2\pi\hbar)^{-1} \frac{\prod_{j=1}^f \hbar\omega_j}{\prod_{j=1}^{f-1} \hbar\omega_j^\ddagger} e^{-\beta V^\ddagger} \\
 &= (2\pi\beta\hbar)^{-1} \frac{\prod_{j=1}^f \beta\hbar\omega_j}{\prod_{j=1}^{f-1} \beta\hbar\omega_j^\ddagger} e^{-\beta V^\ddagger} = (2\pi\beta\hbar)^{-1} \frac{Q^\ddagger(\beta)}{Q_{\text{r}}(\beta)} e^{-\beta V^\ddagger}
 \end{aligned}$$

which is equivalent to classical TST, and shows that the assumptions of RRKM must be strongly related to those of TST.

11.7 Further Reading

The following textbooks and articles will give you further information on the topics of this lecture:

Unimolecular rate theory

- Baron Peters, Reaction Rate Theory and Rare Events (Elsevier), chapter 9 <https://www.sciencedirect.com/book/9780444563491/reaction-rate-theory-and-rare-events-simulations> (available through ETH library using VPN)
- Henriksen and Hansen, Theories of Molecular Reaction Dynamics (Oxford), chapter 7
- W. H. Miller, “Tunneling corrections to unimolecular rate constants, with application to formaldehyde”, *J. Am. Chem. Soc.* **101**, 6810-6814 (1979).